JAMES PATTERSON

Data Engineer

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PROFESSIONAL SUMMARY

Data engineer with three years of experience building and maintaining data pipelines for mid-sized tech companies. I've worked across cloud platforms like AWS and GCP, focusing on ETL processes and data warehouse optimization. I'm most comfortable with Python and SQL but enjoy learning new tools that solve real problems. Outside of work I contribute to open source projects and tinker with home lab infrastructure.

TECHNICAL SKILLS

Languages: Python, SQL, Scala, Bash

Cloud Platforms: AWS (EC2, S3, Lambda, RDS), Google Cloud Platform (BigQuery, Dataflow)

Data Technologies: Apache Spark, Apache Kafka, PostgreSQL, MySQL, MongoDB, Airflow

Tools and Frameworks: Docker, Git, Jenkins, Apache Beam, dbt

Databases: Relational and NoSQL databases, dimensional modeling, database optimization

Other: Linux, API development, data quality frameworks

WORK EXPERIENCE

Data Engineer

DataFlow Systems Ltd

dataflowsystems.io

Liverpool, UK

March 2022 - Present

Designed and deployed Apache Spark pipelines processing 500GB daily across S3 and RDS, reducing query times by 40% through optimized partitioning strategies.

Developed Python ETL scripts using Airflow to orchestrate 30+ data jobs, implementing automated retry logic and error notifications via Slack.

Built real-time Kafka consumers to ingest clickstream data from mobile applications, storing normalized events in PostgreSQL for analytics teams.

Mentored two junior engineers on SQL optimization techniques and data modeling best practices, conducting weekly code reviews for data pipeline PRs.

Junior Data Engineer

Insight Analytics Co

insightanalytics.net

Manchester, UK

July 2021 - February 2022

Created AWS Lambda functions to process and transform data from third-party APIs, automating manual ETL tasks and saving 20 hours weekly.

Wrote comprehensive SQL queries and stored procedures for a 2TB data warehouse, enabling self-service analytics for business intelligence team.

Implemented data validation frameworks using Great Expectations to monitor data quality across 15 source systems and alert on anomalies.

Documented data pipeline architecture and created runbooks for incident response during production data load failures.

Data Analyst

Sterling Financial Services

sterlingfinancial.co.uk

Liverpool, UK

September 2020 - June 2021

Analyzed transaction datasets using Python Pandas and SQL to identify fraud patterns, resulting in 12% reduction in suspicious transactions.

Assisted senior data engineers with AWS RDS migration project, writing test cases and validating data integrity post-migration.

Created automated daily reports using Python scripts and Tableau, eliminating manual spreadsheet creation for stakeholders.

Identified performance bottlenecks in legacy reporting queries and optimized execution time from 45 minutes to 6 minutes through indexing.

EDUCATION

University of Liverpool

BSc Computer Science

Liverpool, UK

September 2017 - June 2020

PROJECTS AND OPEN SOURCE

Real-time Log Analytics Pipeline

github.com/jpatterson92/log-analytics-pipeline

Built Apache Kafka and Spark Streaming application for processing application logs, with visualization dashboard using Grafana.

dbt Data Transformation Suite

github.com/jpatterson92/dbt-transforms

Created reusable dbt models and macros for dimensional modeling, reducing transformation code by 35% across projects.

AWS Cost Optimization Tool

github.com/jpatterson92/aws-cost-monitor

Python utility that analyzes EC2 and RDS usage patterns to recommend cost-saving opportunities for teams.

PostgreSQL Performance Tuning Guide

github.com/jpatterson92/postgres-tuning

Open source documentation and scripts for query optimization and index tuning strategies in PostgreSQL.